Note 3: When μ_X and μ_Y are known, for testing the equality of variances of two normal populations, the test statistic is

$$F = \frac{\frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_X)^2}{\frac{1}{n}\sum_{i=1}^{n}(y_i - \mu_Y)^2} \text{ and follows } F_{m,n}\text{-distribution under } H_0$$

Example 3.1

Two samples of sizes 9 and 8 give the sum of squares of deviations from their respective means as 160 inches square and 91 inches square respectively. Test the hypothesis that the variances of the two populations from which the samples are drawn are equal at 10% level of significance.

Solution:

Step 1 : **Null Hypothesis:** $H_0 : \sigma_X^2 = \sigma_Y^2$

That is there is no significant difference between the two population variances.

Alternative Hypothesis: $H_1 : \sigma_X^2 \neq \sigma_Y^2$

That is there is significant difference between the two population variances.

Step 2 : **Data**

$m = 9, n = 8$

$$\sum_{i=1}^{9}(x_i - \bar{x})^2 = 160 \qquad \sum_{j=1}^{8}(y_j - \bar{y})^2 = 91$$

Step 3 : **Level of significance**

$\alpha = 10\%$

Step 4 : **Test Statistic** $F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{S_1^2}{S_2^2}$, under H_0.

Step 5 : **Calculation**

$$s_X^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2 \text{ and } s_Y^2 = \frac{1}{n-1}\sum_{j=1}^{n}(y_j - \bar{y})^2$$

$$s_X^2 = \frac{160}{8} = 20 \qquad s_Y^2 = \frac{91}{7} = 13$$

$$F_0 = \frac{s_X^2}{s_Y^2} = \frac{20}{13} = 1.54$$

Step 6 : **Critical values**

Since H_1 is a two-sided alternative hypothesis the corresponding critical values are:

$$f_{(8,7),0.05} = 3.73 \text{ and } f_{(8,7),0.95} = \frac{1}{f_{(7,8),0.05}} = \frac{1}{3.5} = 0.286$$

Step 7 : **Decision**

Since $f_{(8,7),0.95} = 0.286 < F_0 = 1.54 < f_{(8,7),0.05} = 3.73$, the null hypothesis is not rejected and we conclude that there is no significant difference between the two population variances.

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